Roll No. | | | | | | | | | | | | | Total No. of Pages : 02

Total No. of Questions : 09

B.Tech. (CE) (Sem.–5)

FORMAL LANGUAGE & AUTOMATA THEORY

Subject Code : BTCS 502-18

M.Code : 91017

Date of Examination : 17-12-22

Time : 3 Hrs. Max. Marks : 60

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains FIVE questions carrying FIVE marks each and students have to attempt any FOUR questions.
3. SECTION-C contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.

SECTION-A

1. Write briefly :

a) Explain in brief about languages and grammars.

b) Define a Regular Expression.

c) Define regular grammar with example.

d) Define context free grammar.

e) What is Greibach normal form? Explain.

f) Define ambiguity in Context free grammar with an example.

g) What are components of a Turing Machine?

h) Define unrestricted grammar.

i) What is context-sensitive grammar? Explain.

j) Define NP-complete problem.

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SECTION-B

2. Differentiate between NFA and DFA.

3. Explain in detail about closure properties of CFLs.

4. Write a detailed note on linear bounded automata.

5. Explain in detail about closure properties of Turing recognizable and Turing-decidable languages

6. Write a detailed note on undecidable problems about languages.

SECTION-C

7. Write a detailed note on the Chomsky hierarchy of languages.

8. a) Explain in detail about the Properties of Regular Expressions.

   b) Write a detailed note on minimization of finite automata.

9. Write a detailed note on Universal Turing Machine.

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.

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